

# Utility of eye lens chemistry to inform the biology of Pacific sleeper sharks



**Old Sharks**



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# NPRB 2301: Ageing of Pacific Sleeper Shark



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**ALASKA BIOMAP**  
scientific services



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# Outline

## NPRB 2301 Objectives

Estimate plausible ranges of age-at-length, age-at-maturity, and lifespan of PSS from 14C.

Investigate sources of potential variation in isotopic values of shark eye lens protein.

Estimate age-related life history parameters and test more robust stock assessment approaches.

# Outline

Species of interest

Eye lens chemistry 101

14C background

Preliminary results

Next steps

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# Species of interest



*Photo Credit: IPHC*



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## Pacific sleeper shark

Incidentally caught in fisheries

Not targeted

Extremely data limited

*Photo Credit: IPHC*

# Species of interest

## Pacific sleeper shark

Incidentally caught in fisheries

Not targeted

Extremely data limited

They get REALLY big

For a great overview of the species:

[Matta et al. 2023 Western Groundfish Conference.](#)

[Matta et al. 2024. Polar Biology article](#)



Picture: Clark et al. 1990

Sugura Bay, Japan

Est size: 660-720 cm TL Matta  
et al. 2024

# Species of interest



## Pacific spiny dogfish

Most common shark in eastern North Pacific

Small shark with large shark life history

Subject of many ageing studies

Used as a proxy species in this study with some bonus analytics

*Photo credit: National Ocean Service*



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# What is eye lens chemistry?

Eyes are like onions, they have layers

Layers sequester chemicals as they form

Provides environmental or biological information at the time of layer formation

Core represents early development, outermost layers is more recent

Can be used to see trends over the lifetime of the animal



# What is eye lens chemistry?

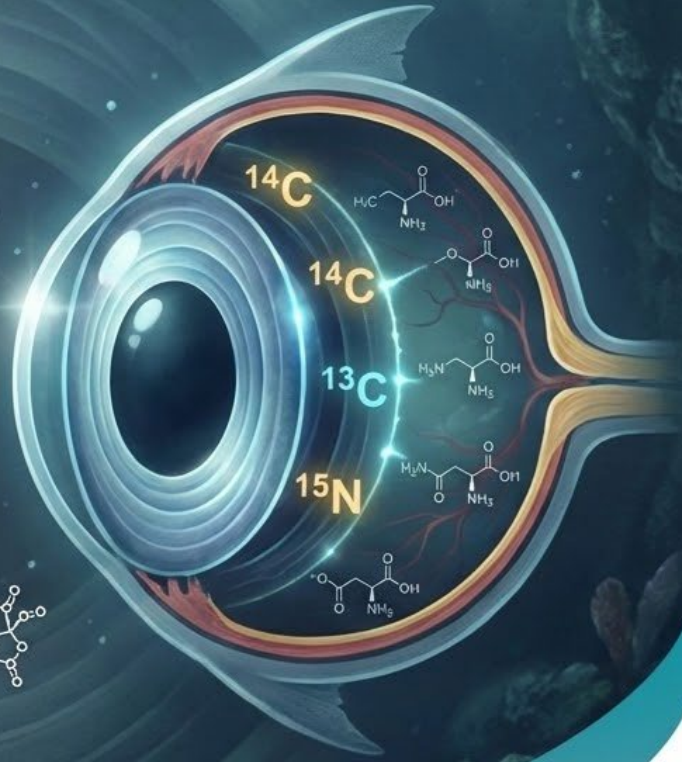
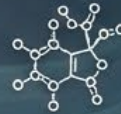
## For the purposes of this study

$^{14}\text{C}$ : bomb produced radiocarbon in the environment as a chronometer

$^{13}\text{C}$ : stable isotope signal of habitat or region

$^{15}\text{N}$ : stable isotope signal of trophic position

CSIA-AA: compound specific isotopic analysis of amino acids, high resolution ecology



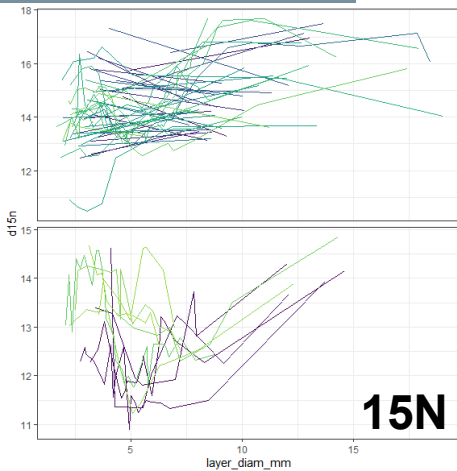
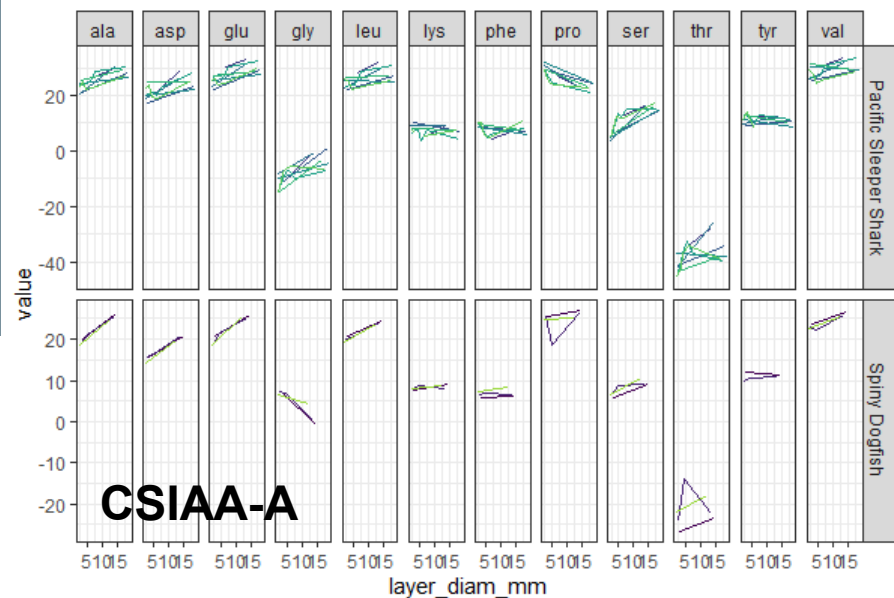
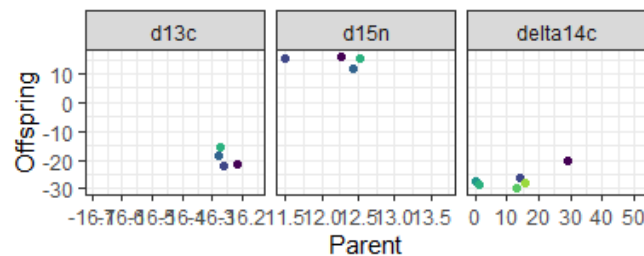


# That's a lot to unpack

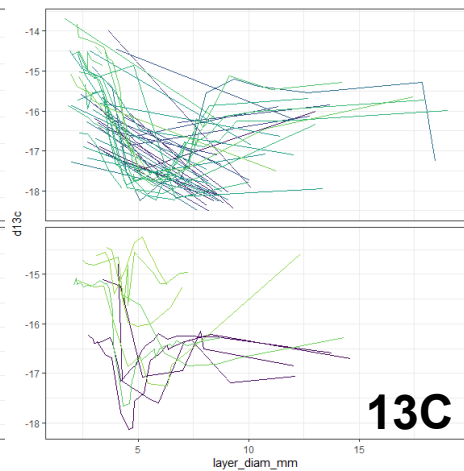
65 Pacific Sleeper Shark, 451 layers

12 Spiny Dogfish, 216 layers

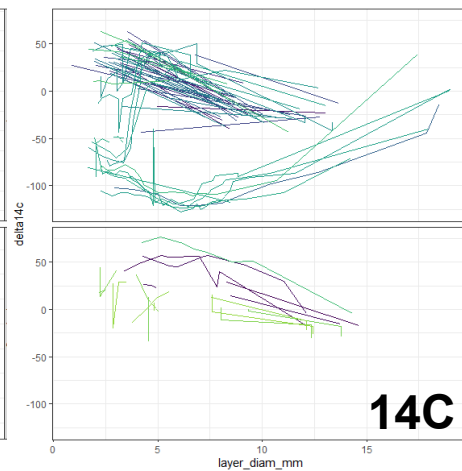
**Maternal**



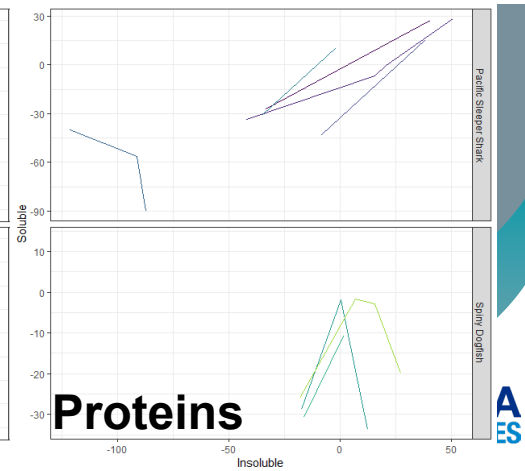
**15N**



**13C**



**14C**



**Proteins**

Pacific Sleeper Shark

Spiny Dogfish

**AES**



# What is eye lens chemistry?

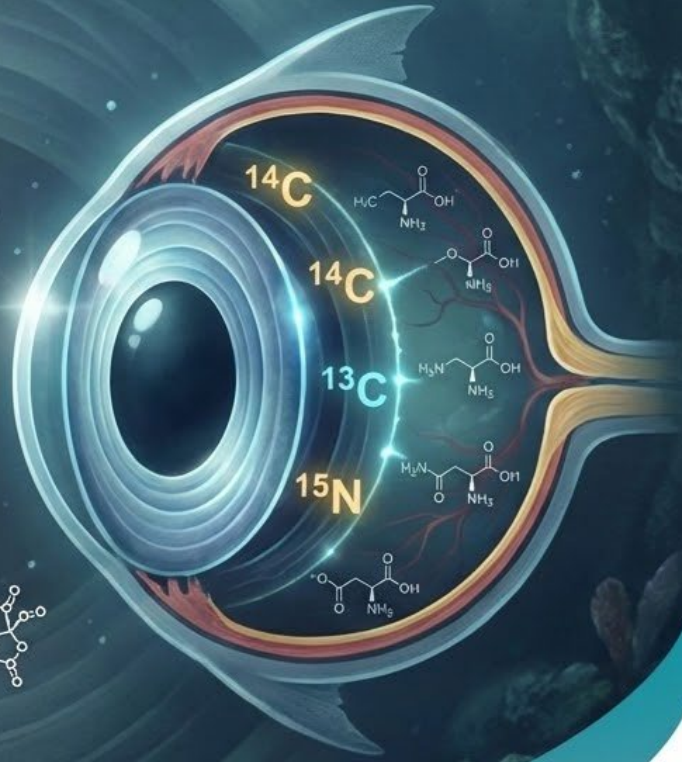
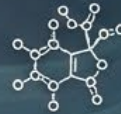
For the purposes of this study ~~TALK~~

14C: bomb produced radiocarbon in the environment as a chronometer

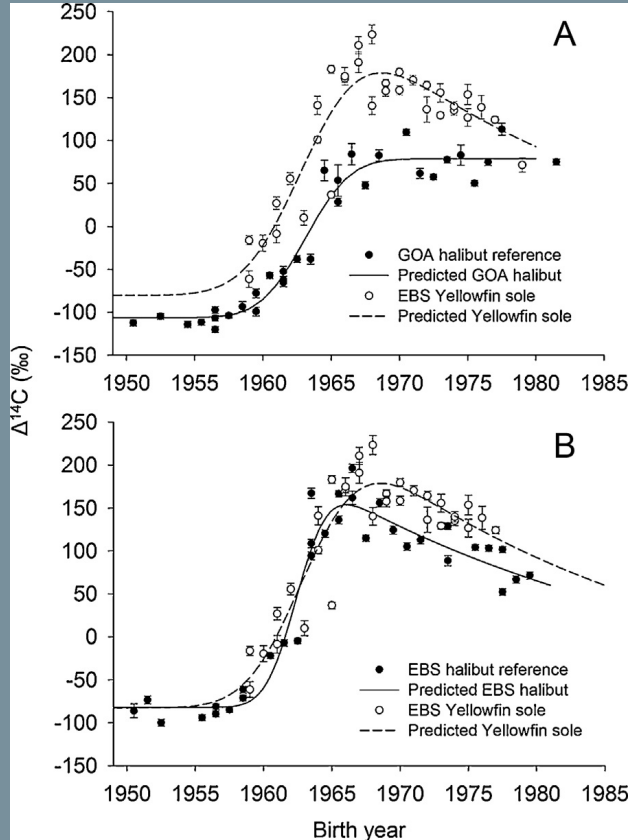
13C: stable isotope signal of habitat or region

15N: stable isotope signal of trophic position

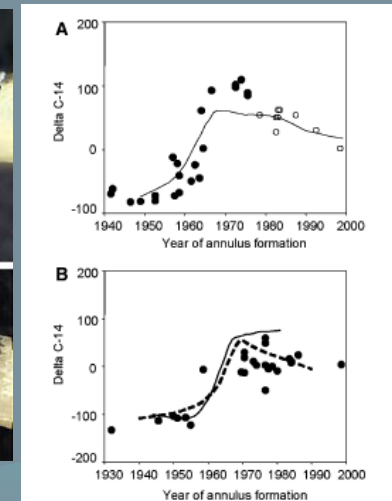
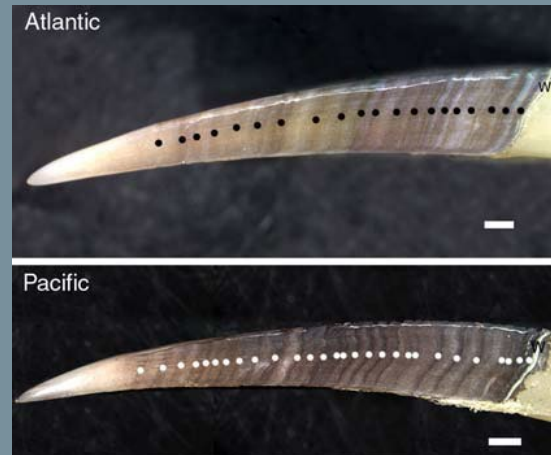
~~CSIA-AA: compound specific isotope analysis of amino acids with high resolution~~



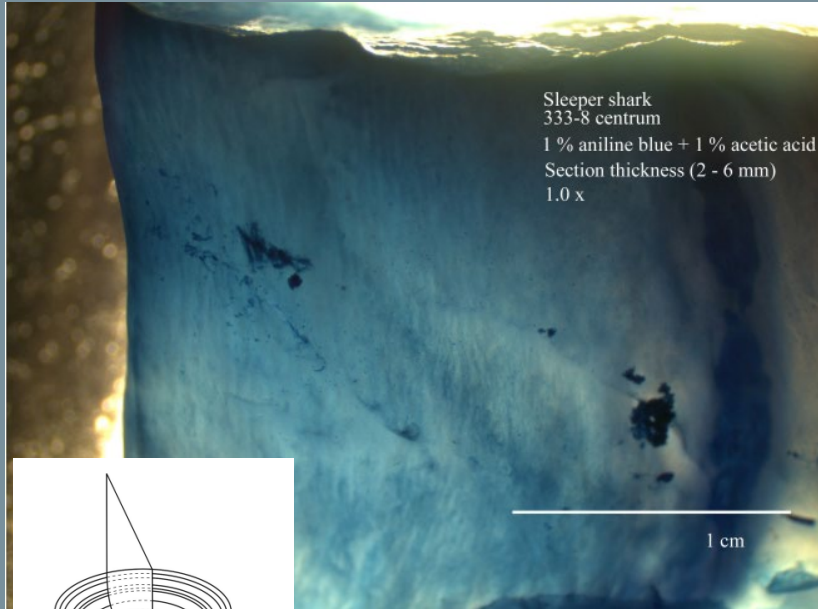
# 14C age validation of teleosts and sharks



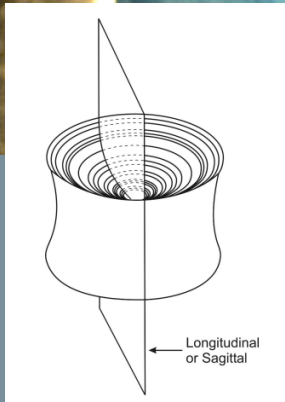
- Often used in teleosts and elasmobranchs to validate ageing methods
- Applied to species which HAVE ageing methods



# 14C for Sleeper Sharks



- Attempts at ageing Somniosids (aka sleeper sharks) with traditional methods have all failed





# $^{14}\text{C}$ for Sleeper Sharks



- Attempts at ageing Somniosids (aka sleeper sharks) with traditional methods have all failed
- Using bomb radiocarbon methods to establish longevity

Nielsen et al. 2016

# 14C for Sleeper Sharks

Are pet clinical trials ready  
for prime time? p. 638

The sense (and antisense)  
of ALS therapy pp. 647 & 708

Keeping fam  
their homes

# Science

\$15  
12 AUGUST  
sciencemag.org

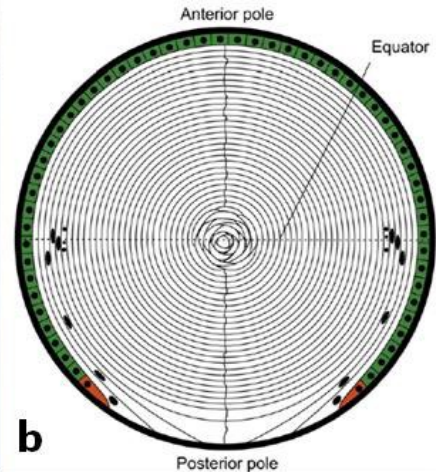


## LONG IN THE TOOTH

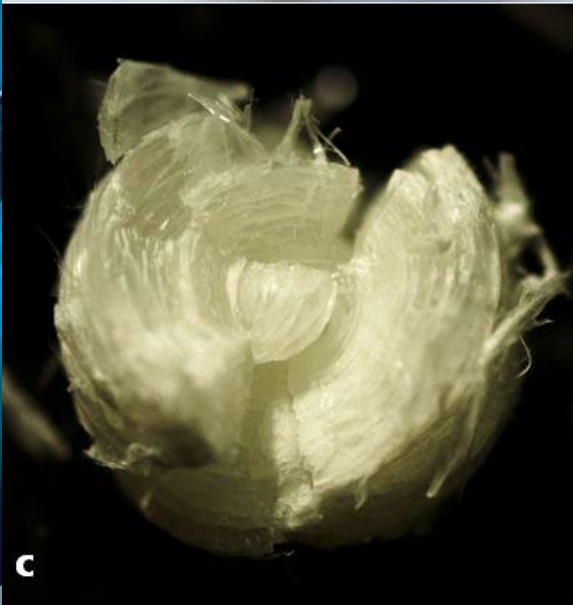
400-year-old sharks in  
Arctic waters p. 702



a



b



c



d



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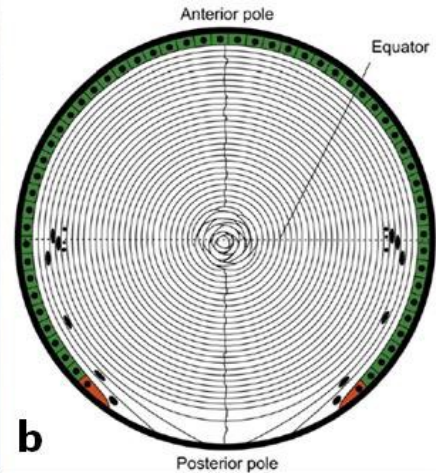


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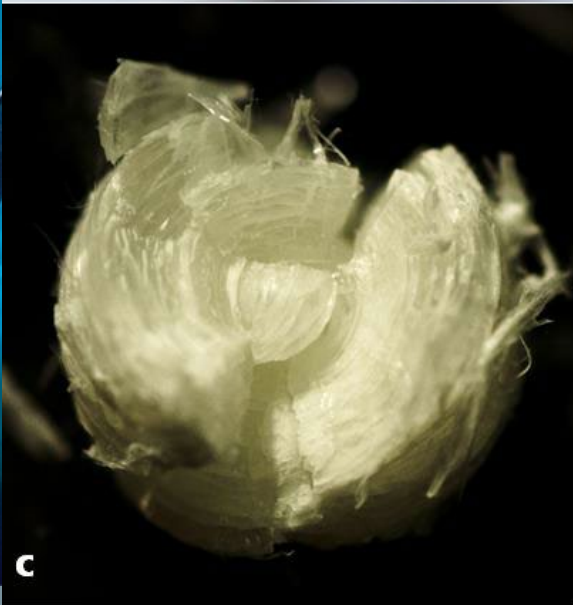
400-year-old sharks in  
Arctic waters p. 702



a



b



c



d



# NPRB 2301: Ageing of Pacific Sleeper Shark

Launching from the lessons learned from Nielsen et al. 2016

- Large sample size
- Lifetime profiles
- 40 - 465 cm TL
- Each eye layer retained for chemical analysis
- $^{14}\text{C}$ ,  $^{13}\text{C}$ ,  $^{15}\text{N}$  and CSIAA-A
- Spiny dogfish with pups for comparison



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# Peeling eye layers

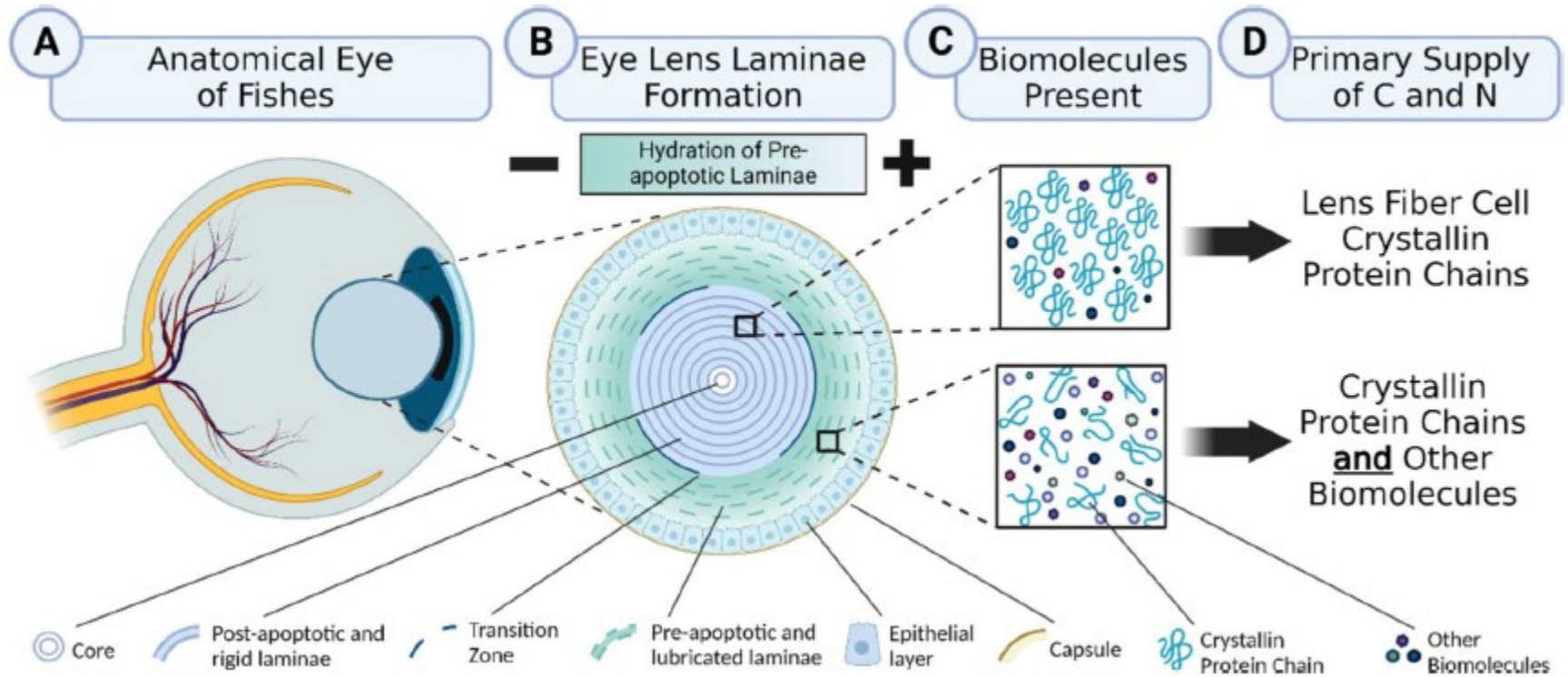
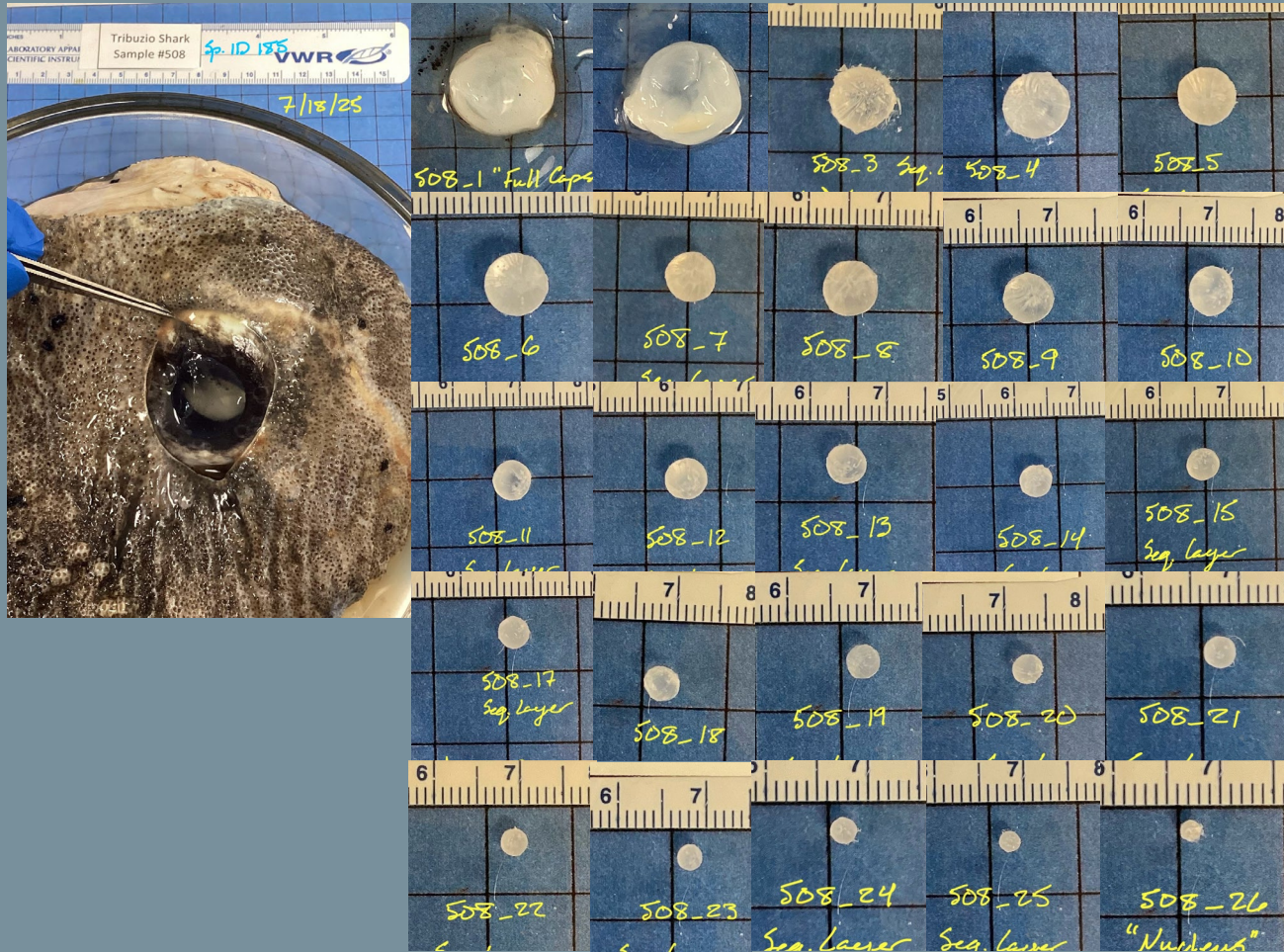


Fig. 1 Kuntz et al. 2025

# Peeling eye layers

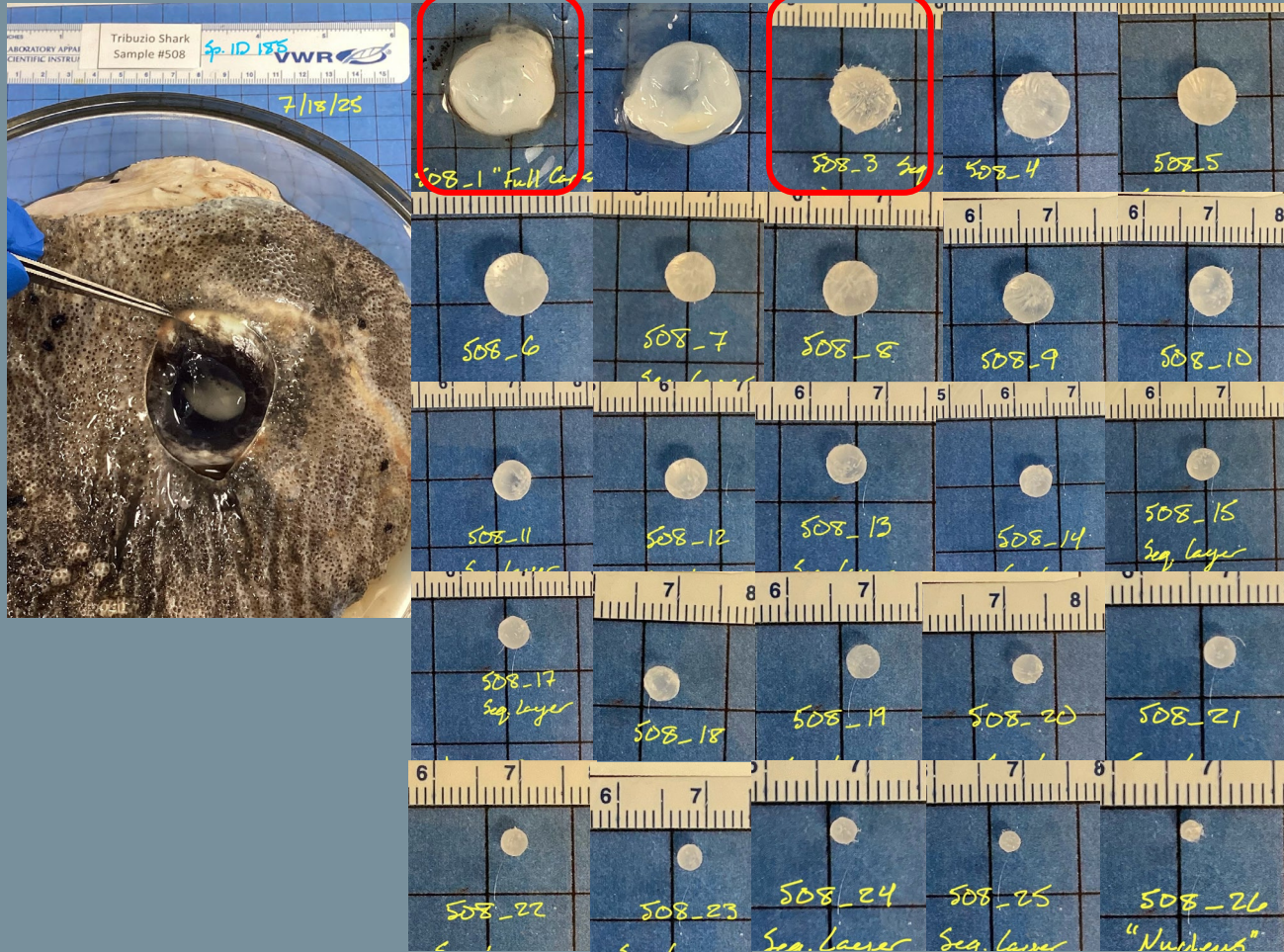


465cm TL Female





# Peeling eye layers



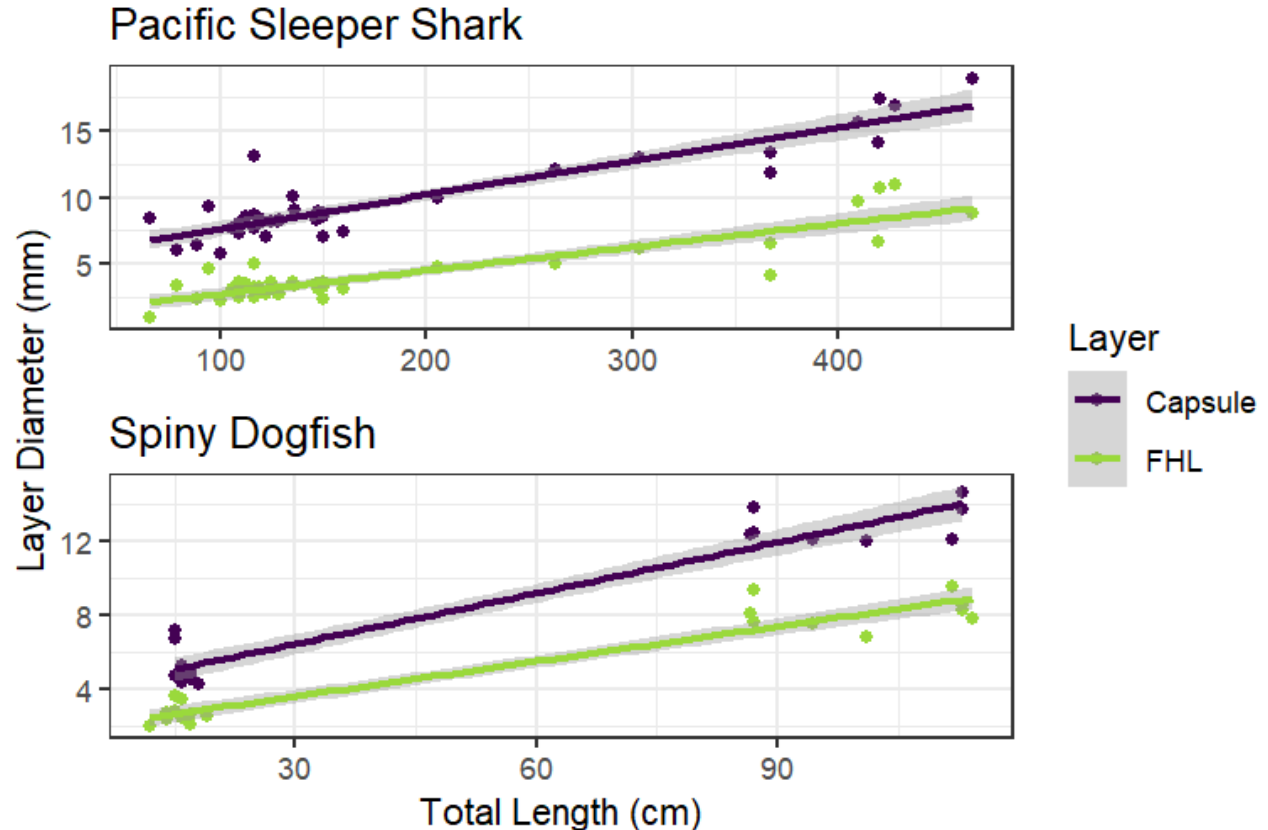
465cm TL Female



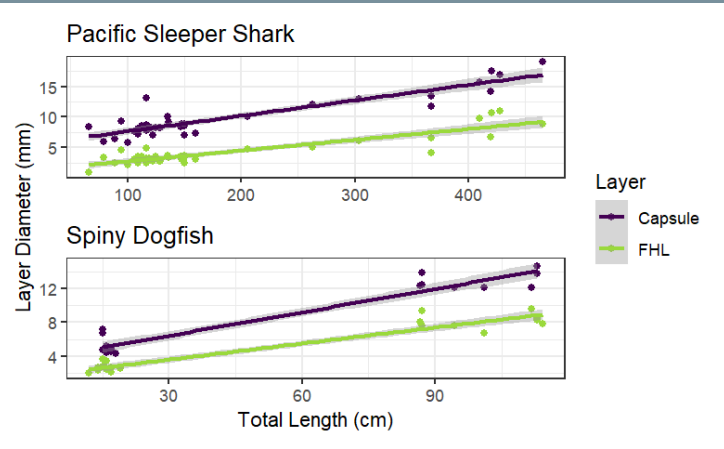
# NPRB 2301: Preliminary Results

Linear relationships

Different slopes



# NPRB 2301: Preliminary Results

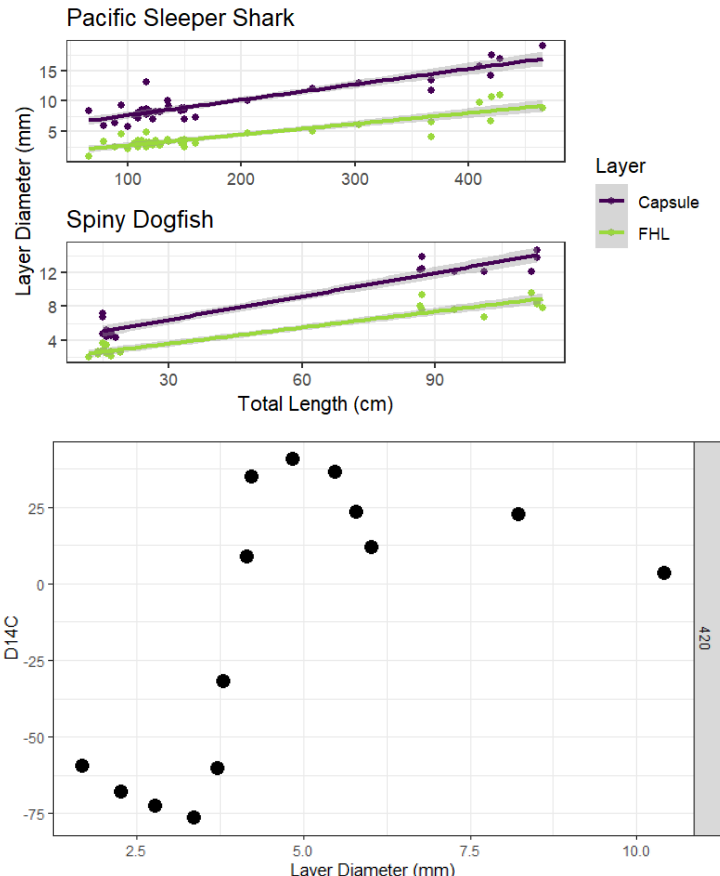


Linear relationship between eye lens diameter and total length

Diameter can be used to estimate length at time of layer formation



# NPRB 2301: Preliminary Results



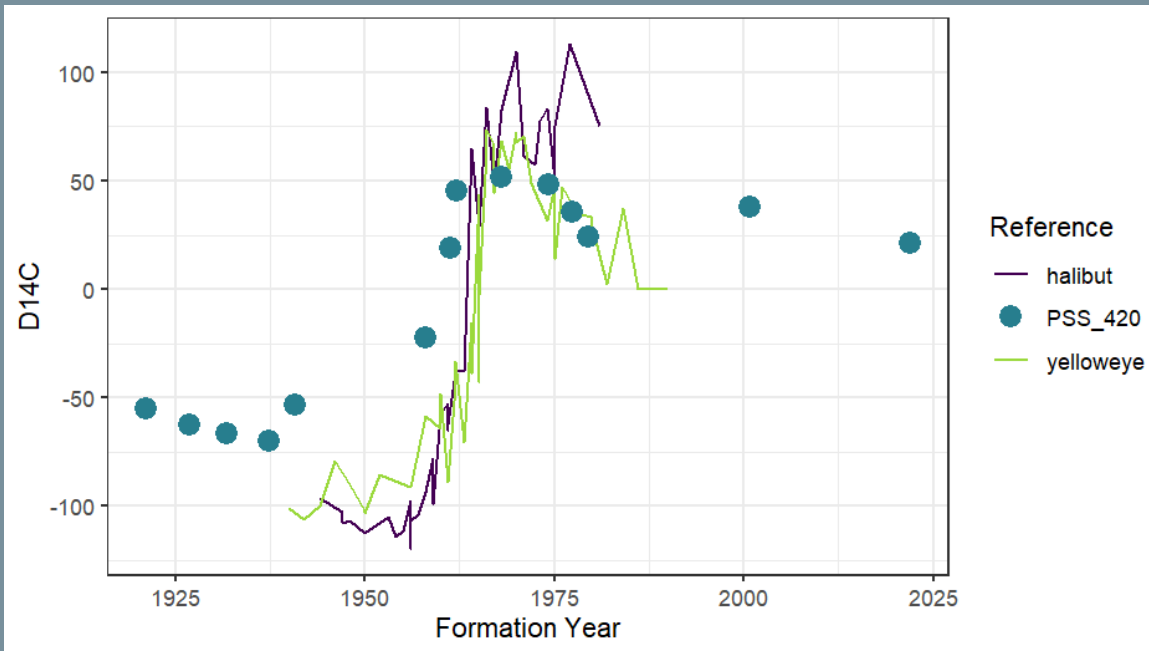
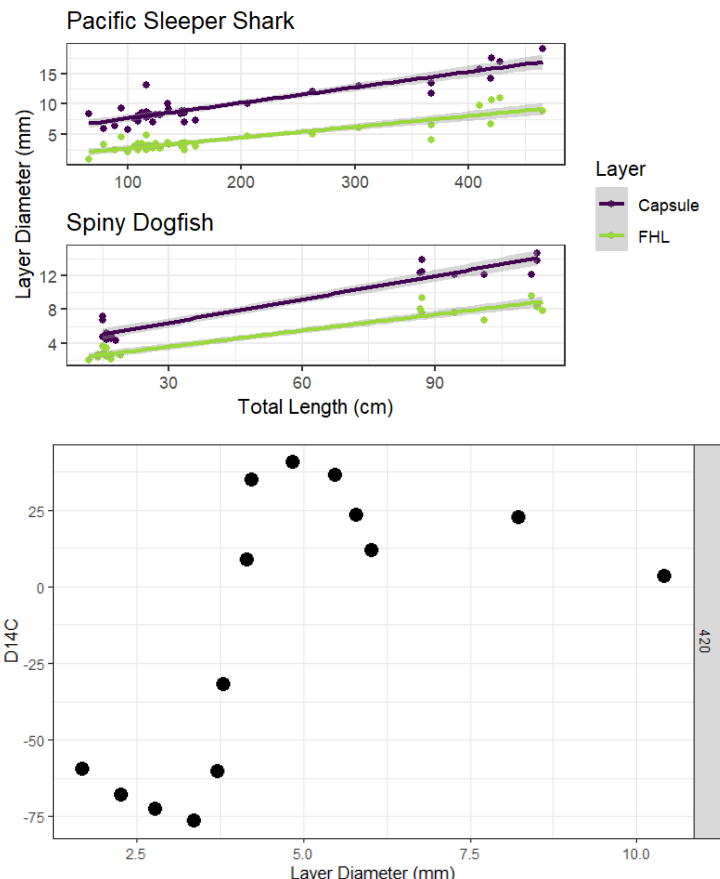
Linear relationship between eye lens diameter and total length

Diameter can be used to estimate length at time of layer formation

Proxy for time

420 cm TL, GOA, 2022

# NPRB 2301: Preliminary Results



420 cm TL, GOA, 2022

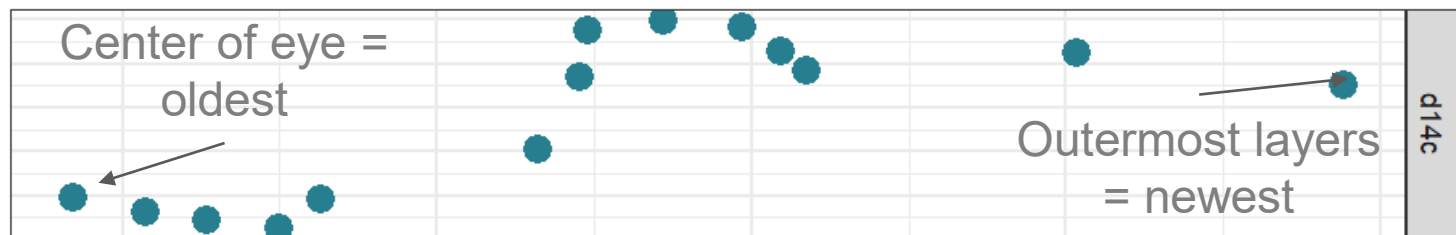


# NPRB 2301: Preliminary Results

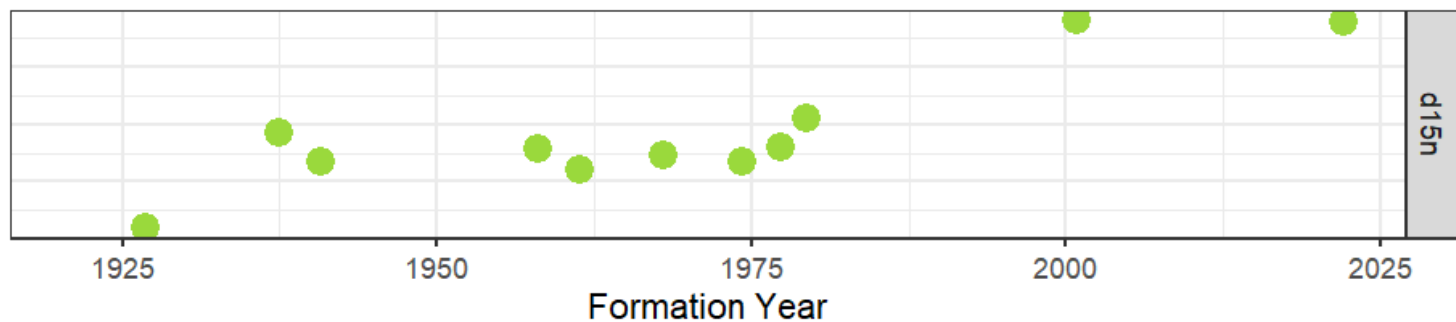
13C indicator  
habitat or region



14C indicator  
of age



15N indicator  
trophic position

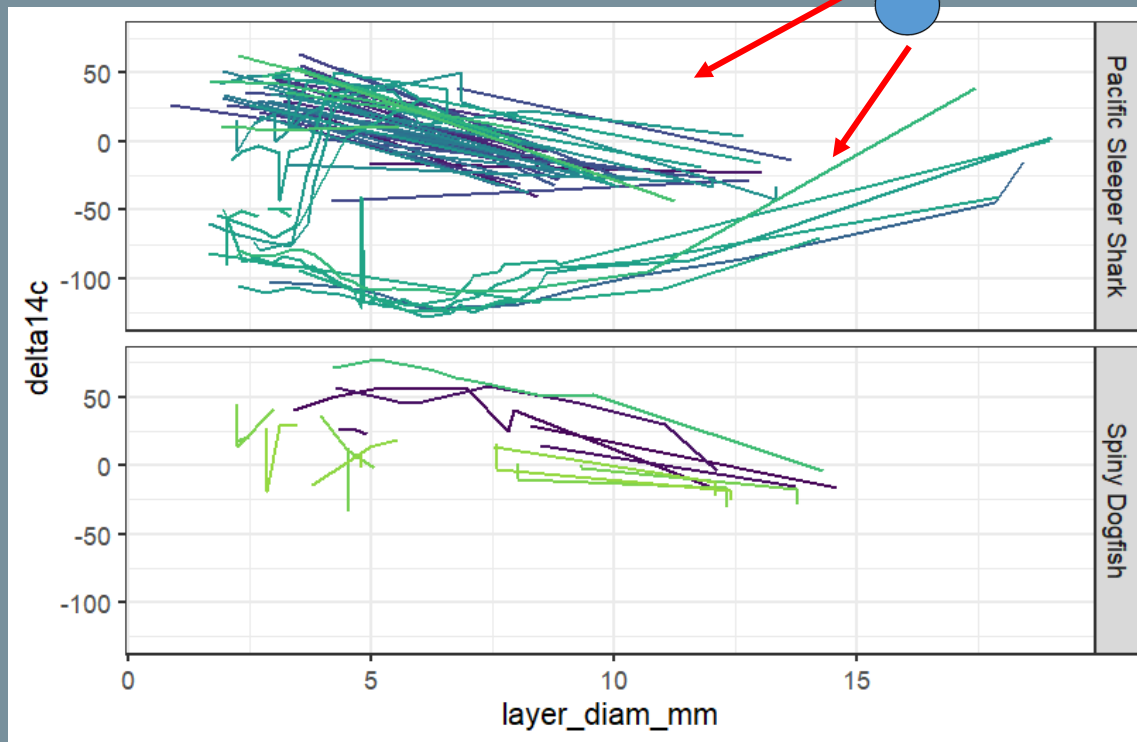






# NPRB 2301: Preliminary Results

Some sample sleuthing  
needed





# NPRB 2301: Next Steps

Some sample sleuthing  
needed

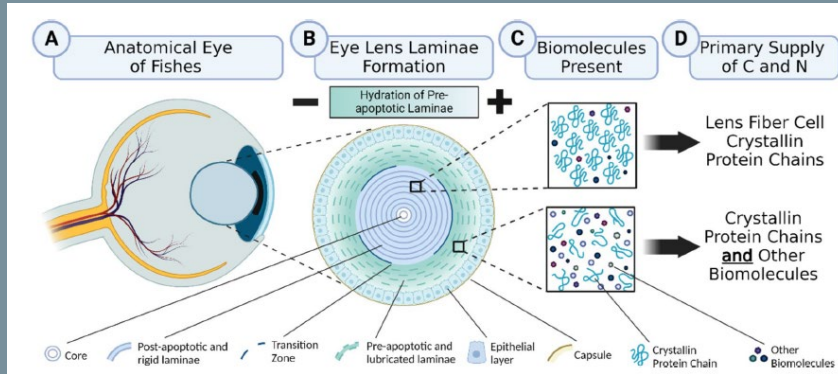
Sequential analytics needed  
first

- Metadata
- Total length estimates
- Protein fractionation

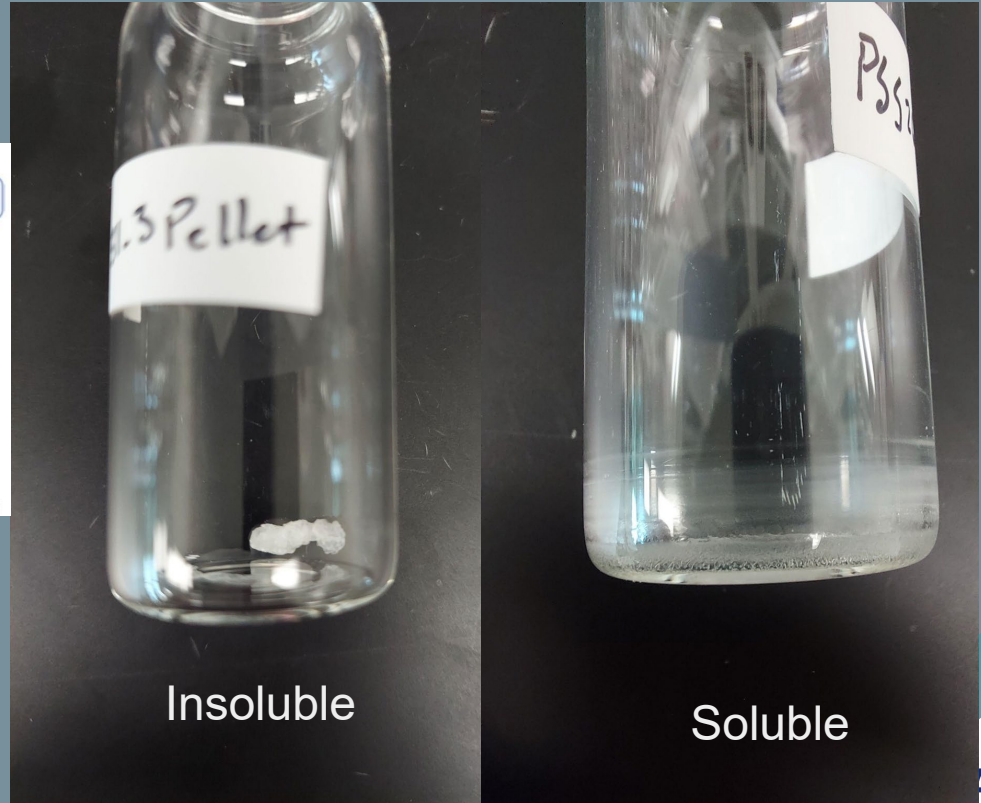


# NPRB 2301: Next Steps

Each layer composed of soluble and insoluble proteins



Modern carbon can infiltrate the soluble protein in the layers, potentially at different rates





# NPRB 2301: Next Steps

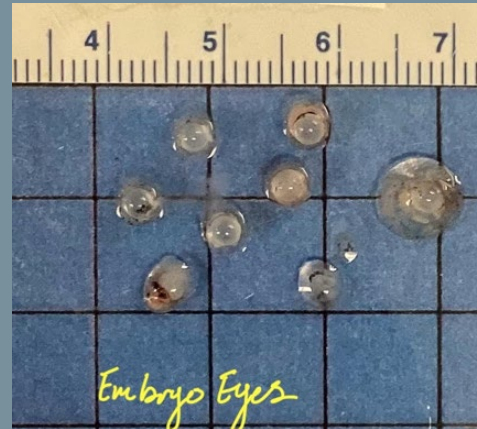
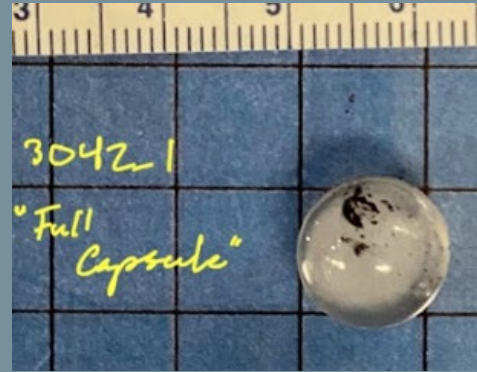
(Obj 2) Finish length estimation and protein fractionation

(Obj 1) Assign year ranges to eye layers

(Obj 1 & 2) Compare/contrast with spiny dogfish and maternal/offspring pairs

(Obj 1 & 2) Interpret 14C, SIA and CSIA-AA results (that's all)

(Obj 3) Feedback to management



# NPRB 2301: Next Steps

Advance the stock assessment  
by informing:



Longevity



Growth rate



Natural mortality



Exploration of expanded  
data-limited assessment  
methods



Demographic simulations



Risk table levels of concern



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# Thanks!!!

North Pacific Research Board for funding and next level support to solve logistical problems

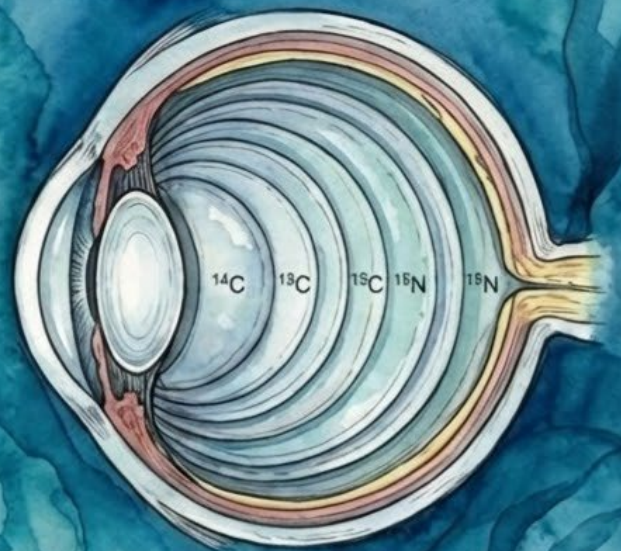
ABL RECA program for lab space and their patience as we figured out our procedures and ABL supervisors for fighting the good fight

All of the at-sea observers, The North Pacific Groundfish Observer Program, and At-Sea Hake Observer Program for supporting the collections

Survey science and vessel crew for taking the time to sample

Fishing vessels who provided samples

Sample processing at UC Santa Cruz Stable Isotope Laboratory, Stanford Stable Isotope Biogeochemistry Laboratory and WHOI NOSAMS



Gemini is trying its hardest, but just can NOT make a sleeper shark

For further questions, contact [cindy.tribuzio@noaa.gov](mailto:cindy.tribuzio@noaa.gov)



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